TRANSCRIPT

Slide 1

Good Morning. I'm Anne. This morning, I will outline my analysis of risk factors related to diabetes from the Pima Indian Diabetes database using descriptive, inferential statistics, and regression analysis.

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Quick roadmap before we start. Section one contains an explanation of the data set and variables. Section two discusses distribution, outliers, and missing values. Sections three to five focus on the inferential findings, including hypothesis testing, correlation, and two regression models. Finally, six recommendations grounded in the findings.

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All analyses and visualisations were produced in R using RStudio.

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The data originated from the National Institute of Diabetes and Digestive and Kidney Diseases and was used by Smith et al. in 1988 to forecast the onset of diabetes in Pima Indian women using the ADAP learning algorithm, an early neural network model. The dataset was accessed via Kaggle, where it was uploaded by Khare in 2022.

The sample consists of 768 women aged 21 years and above living around Phoenix, Arizona. There were 268 diabetic patients in the sample, representing 34.9% of the total and exceeding the world average three to four times according to the International Diabetes Federation's 2021 Atlas. This higher percentage reflects the population's known metabolic risks, which means that models based on such a dataset need to be recalibrated for other populations.

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Before addressing distributions, let us first define a single variable. Eight of the nine variables are physiological measurements that can be understood by everyone: pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, and age, with diabetes diagnoses serving as a binary output. The ninth variable is the Diabetes Pedigree Function, a genetic variable based on family history of diabetes, calculated as a hereditary risk score using a unitless, continuous scale. I will elaborate on this aspect later when discussing linear regression, because it creates a real interpretation problem in the coefficient plot.

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Women mostly fall into the 21 to 25 year category, with fewer numbers as the age group increases. The mean value is greater than the median, indicating a right-skewed distribution. Some outliers, aged 67 and over, are biologically plausible and retained in

the analysis. The IQR runs from 24 to 41, so the middle half of the sample is under 41, which is worth bearing in mind when we look at age as a risk predictor later.

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Glucose has a roughly bell-shaped distribution centred around 117 milligrams per decilitre, with a mean of 121.7 and a standard deviation of 30.5. The spread is wide: the range runs from 44 to 199. Blood pressure is tighter and more symmetrical: mean 72.4, median 72, IQR of 16 millimetres of mercury. Both distributions look reasonable for this population. One thing to keep in mind here is that there are five variables with values of zero: glucose, blood pressure, BMI, skin thickness, and insulin. None of those can be zero in a living person, so I treated them as missing. That drops the regression sample from 768 down to 392, nearly half the data gone. It matters, and I'll return to it in the limitations.

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The BMI distribution is roughly symmetric around the median of 32.3, which already sits in the obese range, by WHO classification. The IQR runs from 27.5 to 36.6, meaning the central half of the sample falls between overweight and obese. The boxplot identifies eight individuals above 50.25 as outliers, with a maximum of 67.1; these are retained as biologically plausible. This is not a healthy-weight sample by any standard definition. Most individuals are already at metabolic risk.

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The bar chart groups pregnancies into four categories: zero, one to two, three to four, and five or more. The largest group is five-plus with 276 women, followed by one to two with 238. 111 women had no pregnancies recorded. When we compare diabetic and non-diabetic women directly, those with diabetes averaged 4.87 pregnancies against 3.30 in those without. Data are right-skewed, so the t-test is replaced with the Wilcoxon rank-sum test. The result is statistically significant: $W = 50{,}985$, $p < 0.001$. Cause-effect direction is uncertain, and I'll address this issue later on Slide 15.

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There are two pairs of variables with correlation in this data set: BMI and skin thickness, which have a correlation of 0.65, as well as insulin and glucose, which have a correlation of 0.58. Neither pair is correlated strongly enough to destabilise the regression; the model can still estimate the outcome reasonably well, but it is unable to separate their individual contributions. This means that the BMI coefficient includes some effect of the skin thickness and vice versa. The standard errors understate that uncertainty, which is something to bear in mind when reading the coefficient estimates in the linear regression. It is not a fatal flaw, but it is a real one.

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The relationship between BMI category and diabetes rate is obvious. The underweight group, 13.3%, is bigger than the normal-weight group, 6.9%, which is probably a consequence of the small sample size and potential confounding by older or sick patients. It does not affect the general trend. The relationship is statistically significant: chi-square = 78.25, $p < 0.001$.

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Risk of developing diabetes grows with age: 21.2% of under-thirties against 55.1% of 40 to 49-year-olds, dropping to 48.3% in those 50 and above. The plateau between the last two groups is worth noting; a linear age term in regression is only an approximation of what is clearly a non-linear pattern. One possible explanation for the drop is survivorship bias: those with severe, undiagnosed diabetes may have already left the sample through mortality or withdrawal, leaving a relatively healthier subset. This is a hypothesis that the cross-sectional design cannot confirm or rule out.

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It is obvious that there is a difference in the mean value of glucose in the two groups. It is 142.3 milligrams per decilitre in the group of patients with diabetes, while in those who don’t have the disease, the level is 110.6. The difference of 31.7 is not only statistically significant but also clinically significant, as the fasting glucose threshold for

diabetes is 126 milligrams per decilitre, as defined by WHO. To prove my hypothesis, I ran both Welch's t-test and the Wilcoxon rank-sum test. The Q-Q plots show acceptable normality in the middle range but some deviation in the upper tail, so I wanted parametric and non-parametric confirmation rather than relying on a single test. Both return p-values below 2.2 × 10^-16.

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The percentage of pregnancies among patients with diabetes is consistent with the expected percentages: 20.2% have one to two pregnancies, while 47.8% have five or more pregnancies. However, the group that does not have any pregnancies is the outlier because 34.2% of diabetics do not have any pregnancies. The most convincing reason for this is reverse causality, since undetected insulin resistance affects fertility and does not imply a lower metabolic risk associated with fewer pregnancies. As Dabelea et al. argued, a longitudinal design is needed to properly separate metabolic and reproductive pathways, something cross-sectional data cannot establish.

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Mean glucose levels increase with age, but not uniformly. They are 114.6 in individuals below 30 years old, 126.1 in 30–39 year olds, 125.7 in 40–49 year olds, and 139.6 in

individuals above 50 years old. ANOVA proves that there is a significant difference here: $F = 20.44$, $p < 0.001$. Kruskal-Wallis analysis was also used as a nonparametric robustness check. Tukey tests show that the two middle-aged groups do not differ from each other but differ from the youngest and oldest groups. There are high outliers in the under-30 group, which might be explained by early stages of insulin resistance, consistent with this population's metabolic profile.

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A multiple linear regression model with 392 complete observations reveals that 39% of the variance in glucose can be explained by the model: $F(7, 384) = 36.76$, $p < 0.001$. Three predictors are significant at the 5% level: insulin, age, and blood pressure, all positive. Diabetes pedigree function has the largest raw coefficient of roughly 4.2 and appears to be dominant in the plot, although that is merely an artefact of different units; since it is an index, there is no way to compare it to physiological values. The remaining 60% is likely from diet, physical activity, and HbA1c. Insulin is excluded from the logistic model due to its high correlation with glucose, $r = 0.58$.

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The diagnostic plots are worth working through because they qualify the model's conclusions in ways that matter. The residuals-versus-fitted plot shows a mild systematic curve rather than a flat scatter around zero, suggesting the linear specification may be missing something: possibly a non-linear glucose trajectory, or an

interaction the additive model does not capture. The Q-Q plot is well-behaved across the middle range but deviates in the upper tail, where high-glucose cases pull away from the theoretical line. Approximate normality holds well enough for inference, but that tail behaviour is a flag. The scale-location plot is the most consequential finding. There is a visible upward trend in the spread of residuals as fitted values increase; the errors grow as predicted glucose rises. Practically, this means the model's confidence intervals become increasingly unreliable at higher predicted glucose values, precisely where the diabetic cases concentrate. Standard errors are probably underestimated in that region. Finally, observations 226 and 5 show elevated leverage, though neither breaches Cook's distance, so neither is distorting the overall fit. I'll return to this in the limitations.

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This model was built to identify predictors of glucose, not as a clinical tool. With that purpose in mind, 39% explained variance is meaningful, though the scatter confirms that glucose variation here is not fully captured by these seven variables.

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The three-predictor logistic model uses BMI, age, and glucose on 752 complete cases. All three are significant: BMI at 1.09, age at 1.03, and glucose at 1.04. Age is significant at $p < 0.001$ here; that changes on the next slide.

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Incorporating the number of pregnancies makes a difference. Pregnancies have the highest impact, with an odds ratio of 1.12, meaning each additional pregnancy raises the odds by 12%. BMI stays at 1.10, glucose at 1.04. However, age drops to 1.01 and becomes insignificant, $p = 0.194$. After controlling for pregnancy history, age stops having additional influence; the age effect in the simpler model appears partly to be a proxy for reproductive history.

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The scatterplot makes a point that the regression coefficients alone cannot illustrate as clearly. For any given BMI value, the diabetic cases cluster consistently higher on glucose than the non-diabetic cases. The two trend lines run parallel and do not intersect across the observed range. If glucose and BMI were simply measuring the same underlying thing, knowing one would largely explain away the other, and a higher BMI would account for the glucose difference. It does not. There are individuals with identical BMI values sitting on opposite sides of the diagnostic boundary, separated by their glucose level. That is the empirical case for including both variables in any

screening instrument: not a modelling convention, but because the data shows they are doing different work.

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The interaction model has a higher AIC of 724.3 against 722.9 without it. The likelihood ratio test produces a deviance change of 0.564 with a p-value of 0.453, while the odds ratio of the interaction term equals 1.001, close to 1. Formally, interaction is insignificant. However, it is seen from the graph of predicted probabilities that there is a visibly widening separation above BMI 35. As noted by van de Schoot et al. in 2021, Bayesian methods handle this structural uncertainty more explicitly. I believe that it is a power limitation rather than a true null.

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At the 0.50 threshold, the accuracy is 77%, and the specificity 87.9%, but the sensitivity is only 56.8%. The Hosmer-Lemeshow test returns p = 0.197, an adequate fit per Hosmer et al. However, statistical fitness and clinical usefulness are different things. Sensitivity of 56.8% means more than four in ten diabetics are missed, 114 false negatives against 59 false positives, roughly a 2:1 ratio. In a screening context, this is the clinical cost of defaulting to the 0.50 threshold, not a statistical footnote.

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Two types of signal emerge from the analysis. The metabolic signal, glucose and BMI, is consistent, independent, and significant at every analytical stage from the chi-square tests through both logistic models. The life-course signal, carried by age and pregnancy history, is statistically significant but neither modifiable nor causally established from this cross-sectional design. The distinction matters practically: glucose and BMI respond to dietary change, physical activity, and medical management; age and pregnancy count are contextual markers, informative, but not something you can intervene on. Insulin occupies a third category. It was the strongest predictor of glucose in the linear model, pointing to the metabolic mechanism linking the two variables, but it was excluded from the logistic regression because of its strong correlation with glucose at r equals 0.58, which would introduce redundancy.

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I want to highlight three limitations that cannot be considered caveats. Firstly, there is missing data. Since zeroes were removed from five variables, the size of the regression data decreased from 768 observations to 392. Having a biological justification for excluding zeros does not resolve the statistical concern: the key question is whether those missing values were missing completely at random. If patients with more serious problems whose conditions were left untreated were characterised by missing values,

then 392 observations would represent a healthier group, and optimistic bias could be expected in the results, which we cannot quantify after the fact.

Secondly, according to the scale-location plot, standard errors at high glucose values are probably underestimated, and associated confidence intervals should be taken into account with care.

Thirdly, all classification metrics were calculated on the same data used to fit the model: no train-test split, no cross-validation. The 77% accuracy, 56.8% sensitivity, and 87.9% specificity are in-sample estimates and almost certainly optimistic.

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Here are six suggestions. Blood glucose and BMI should be mandatory inputs in any screening tool, consistent, independent, and non-redundant. The 0.50 threshold needs to be reconsidered; the range from 0.30 to 0.35 would recover many of the 114 missed diagnoses, with the cutoff set by the clinical cost of false negatives. Age and pregnancy status are informative context markers, but unmodifiable, and their significance shifts with model specification. Multiple imputation should replace complete-case analysis. At a minimum, cross-validation within the sample is needed before any clinical application. And Bayesian logistic regression, as van de Schoot et al. described, would allow informative priors from prior Pima studies to be incorporated and handle class imbalance more explicitly.

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This dataset is small, single-population, and cross-sectional. And yet it tells a coherent story. Glucose and BMI stand out as consistent, independent signals of diabetes risk, appearing in the subgroup charts, in the hypothesis tests, and in both logistic models. The logistic regression is adequate for population-level risk stratification. But adequacy for population stratification is not the same thing as readiness for individual clinical use. A model that misses 114 diabetics, a 56.8% sensitivity at the 0.50 threshold, is not a screening tool. That gap between what the statistics say and what the model would actually do in a clinic is precisely what the six recommendations are designed to close. Lowering the threshold, applying multiple imputation, running cross-validation, and extending to Bayesian methods are the steps between a well-conducted analysis and something you could responsibly deploy. Thank you.

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